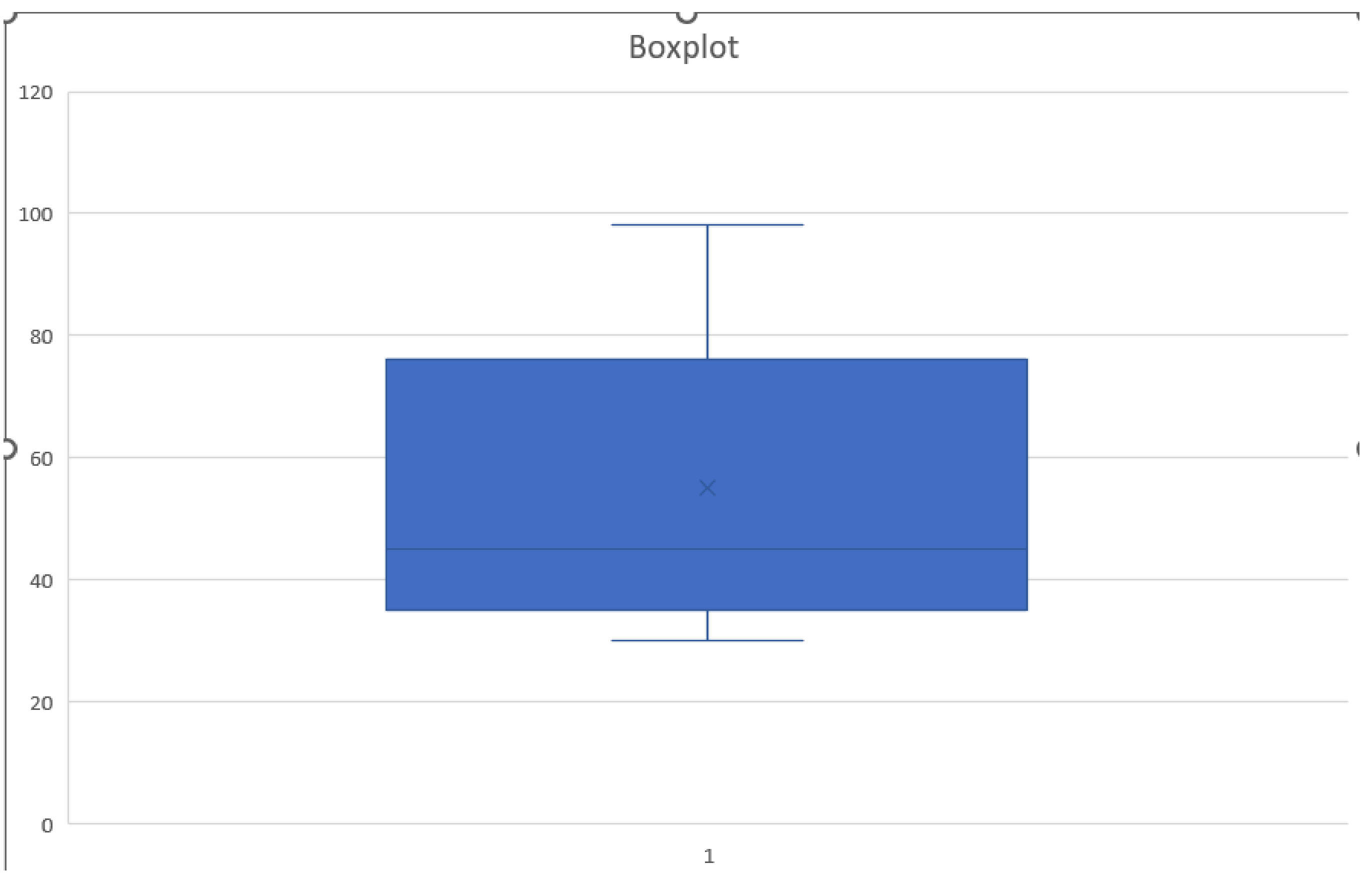
Lab 2

Questions

Given the following measures on the Statistics scores for a group of 25 second year's students:

The smallest 3 values	The first quartile	The median	The third quartile	The largest 3 values
30, 33, 37	40	45	60	72, 80, 98

1) Draw the Box-Plot.



2) Find a measure of central tendency

Median = 45 (كده كده معطى)

3) Is the value "15" an outlier? Why?

IQR=Q3−Q1=60−40=20
Lower limit=Q1−1.5×IQR=40−30=10
Upper limit =Q3+1.5×IQR=60+30=90
→ NO , 15 not an Outlier since 15 > 10 (lower limit)

4) Determine the direction of the skewness (without calculation).

Positively skewed (right-skewed)

5) Calculate the coefficient of skewness, and comment.

β = ((Q3 −Q2) − (Q2 −Q1)) / (Q3 −Q1)

β = ((60 - 45) - (45-40)) / (60-40) = 0.5

β = 0.5 ⇒ Slight positive skewness

6) Is it a difficult exam? Why?

Yes, it seems difficult, because the majority of scores are concentrated in the lower half of the scale, while only few students scored high (98, 80, 72).

Some Exercises

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(1) What name is given to a table that lists all the values that a discrete random variable X can assume and their corresponding probbilities?

Probability Distribution Table

(2) For the probability distribution of a discrete random variable, the probability of any single value of X is always

a. In the range 0 to 1 b. 1.0 c. less than zero

answer → b (in the range 0 to 1)

(3) For the probability distribution of a discrete random variable, the sum of the probabilities is always ....

answer = 1

(4) The parameters of the binomial probability distribution are .... and ....

(5) The binomial distribution is skewed to the right if π 0.5

(6) The parameter/ parameters of the Poisson probability distribution is/ are ....

4- **n**: Number of trials

- **π**:Probability of success in each trial.
  Answer = n and π.
  5 - not equal
  6- **λ** (mean rate of occurrence).

(7) Find the mean and the standard deviation from the followin table

x	-4	0	1	2
$P(X = x)$	0.2	0.3	0.3	0.2

- **Mean (μ)**:
  μ=∑x⋅P(X=x)=(−4×0.2)+(0×0.3)+(1×0.3)+(2×0.2)=−0.3
- **Variance (σ2)**:
  σ2=∑(x−μ)2⋅P(X=x)=4.41
- **Standard Deviation (σ)**:
  σ=2.1

(8) A factory has eight machines. The probability is 0.04 that any machine will break down at any time. Find the probability that at any given time:

a. All eight machines will be broken down

b. Exactly two machines will be broken down

c. None of the machines will be broken down

$$P(X = x) = \binom{n}{x} p^x.(1-p)^{n-x} \;;$$
$$\text{Or} = (C_x^n)\, p^x.(1-p)^{n-x}$$

a. Probability that **all 8 machines** break down

P(X=8)=(0.04)8≈0

b. Probability exactly 2 break down (Binomial formula):

P(X=2)=(8 2)(0.04)2(0.96)6≈0.042

c. Probability none break down:

P(X=0)=(0.96)8≈0.721

(10) A high school boys' basketball team averages 1.2 technical fouls per game. Find the probability that in agiven game this team will commit:

a. Exactly three technical fouls

b. At least two technical fouls

c. Find the mean and the standard deviation

$$P(X) = \frac{e^{-\lambda}.\lambda^x}{x!} \qquad , \lambda > 0 , \qquad x = 0,1,2,3,\dots\dots$$

a. P(X=3)

P(X=3)=(e pow(-1.2) * 1.2 pow(3)) / 3! ≈0.0867

b. P(X≥2)

P(X≥2)=1−P(X=0)−P(X=1)≈0.337

c. Mean (μ) = **1.2**

Standard Deviation (σ) = sqrt(1.2)≈1.095